

ZipDepth Backbone for YOLO26n: Follow-up Experiments

Lab entry task · Amirmahdi · October 2026 · KITTI 2D detection, Colab T4

Background. YOLO26n (**A**) reaches 0.588 mAP50-95 on KITTI val. Replacing its backbone with the depth-pretrained ZipDepth-base encoder plus 1×1 adapters (**B**) gives 0.472. Three explanations of this gap were tested, each with one change to Run B:

- **Adapter capacity (B-A3).** A single 1×1 layer is only a linear projection and may be too weak to turn depth features into detection features. B-A3 uses 3-layer adapters (1×1, 3×3, 1×1 Conv-BN-SiLU).
- **Decoder initialisation (B-RN).** The COCO-pretrained neck and head were trained on YOLO backbone features and may not suit ZipDepth features. B-RN trains the neck and head from random initialisation.
- **Training length.** Run B had not converged after 50 epochs. A, B and B-A3 were continued for up to 20 more epochs from their best weights with early stopping (lr 1e-3 to 1e-4, no mosaic, patience 5).

All other settings are identical to the original runs (data, 640 px, batch 16, SGD, seed 0, evaluation code), and every new model passed the sanity checks (incl. overfit and resume tests) before training.

Results

Run	Adapter	Neck/head	mAP50-95 50 ep	mAP50-95 70 ep	mAP50 70 ep	small car 70 ep	Params (fused)	GFLOPs	img/s b32
A	none	COCO	0.588	0.602	0.863	0.563	2.38 M	5.3	419
B	1 layer	COCO	0.472	0.490	0.748	0.479	7.14 M	17.8	288
B-RN	1 layer	random	0.461	not run	not run	not run	7.14 M	17.8	≈ B
B-A3	3 layers	COCO	0.512	0.531	0.802	0.530	8.13 M	21.0	270

Table 1. Validation set (1,496 images, 8,128 objects). '70 ep' is the best checkpoint after the 20-epoch continuation. Throughput: batch 32, FP16, measured back to back in one session.

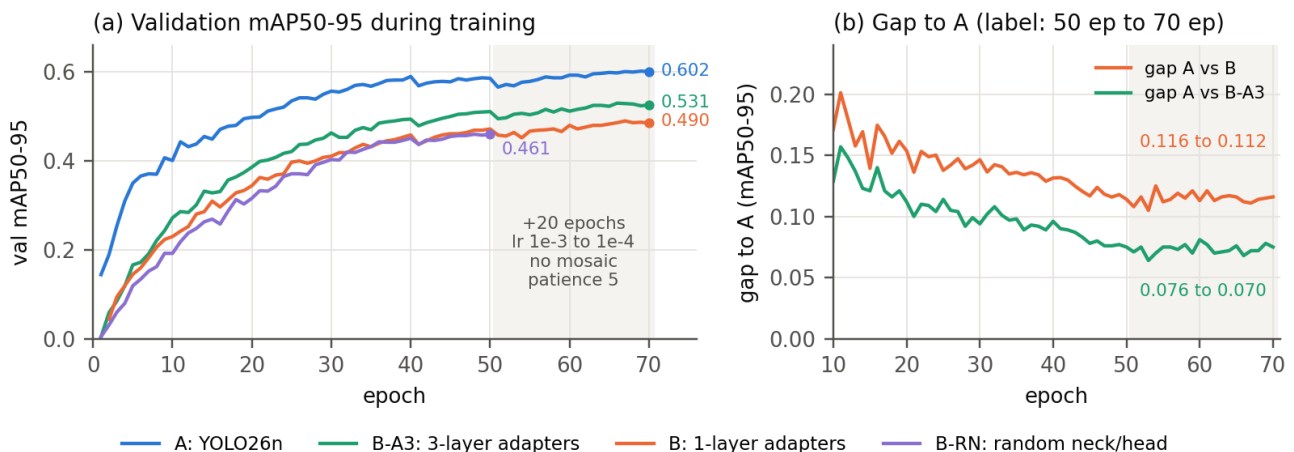


Figure 1. (a) Per-epoch validation mAP50-95 (training-time validation; Run B's first 50 epochs were read from the notebook plot, ± 0.006). Labels give the final evaluation of each best checkpoint. (b) Gap between A and each ZipDepth model; it stops shrinking once A converges and stays flat during the continuation.

Can longer training close the gap to YOLO26n?

No. The ZipDepth models learn too slowly and have already stopped improving.

- **Equal gains:** in the 20 continuation epochs A improved by 0.012, B by 0.018 and B-A3 by 0.020, so the gap to A shrank by only **0.006** for B-A3 (from 0.076 to 0.070) and 0.004 for B (from 0.116 to 0.112).
- **Plateau:** B-A3 gained 0.0017 mAP per epoch at the end of the first 50 epochs but lost 0.0012 per epoch over the last 5 continuation epochs; B also turned slightly downward (0.0004 per epoch), and B-RN gained only 0.0008 per epoch at epoch 50. A is still rising by 0.0007 per epoch.
- **Extrapolation:** at the narrowing rate actually measured (about 0.0003 mAP per epoch), closing the remaining 0.070 would take more than 200 further epochs; even B-A3's best observed rate, with A frozen, would need more than 40. The flat curves in Figure 1 support neither.
- Early stopping (patience 5) did not trigger, but the best epochs (16 to 19 of 20) and the flat curves show that the remaining gains are close to zero.

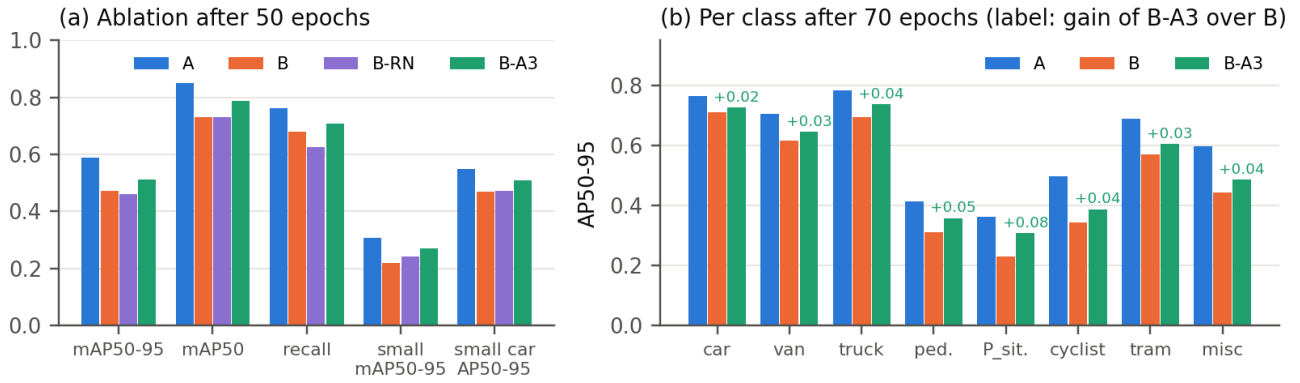


Figure 2. (a) Adapter depth and neck/head initialisation after the common 50-epoch schedule. (b) AP50-95 per class after the continuation; labels give the gain of the 3-layer adapters over B.

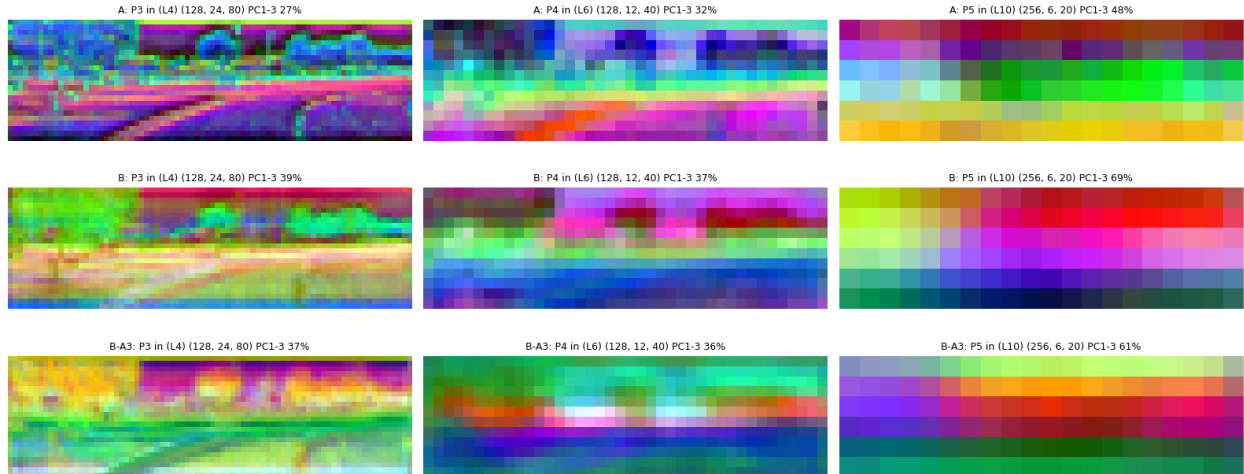


Figure 3. PCA-RGB of the backbone outputs handed to the neck (P3 / P4 / P5) on the same val image, after the continuation; rows A, B, B-A3. Titles give the variance explained by three components.

Findings

- **Adapter capacity matters most:** 3-layer adapters add 0.040 mAP50-95 after 50 epochs and 0.041 after 70. They help all 8 classes (most: Person_sitting 0.08, pedestrian 0.05) and small objects (small mAP50-95 0.067, small cars 0.051), for 1 M extra parameters and 3.2 extra GFLOPs. The 3×3 layer adds spatial context and non-linearity that a 1×1 projection cannot provide. In Figure 3, B-A3's P5 is less dominated by the smooth depth-like gradient (61 % of variance in three components, compared with 69 % for B and 48 % for A).
- **Decoder initialisation does not matter:** B-RN is 0.011 below B, within single-seed noise, and has plateaued; it trades recall for precision (R 0.627 vs 0.679, P 0.836 vs 0.750). COCO initialisation of the neck/head is worth only about 0.01, so the gap to A comes from the backbone features, not from the decoder.
- **Ranking:** A > B-A3 > B ≈ B-RN. The best ZipDepth model is still 0.070 mAP50-95 below YOLO26n while using 3.4× the parameters, 3.9× the FLOPs and 0.65× the throughput.
- **No advantage for distant objects:** compared with A, B-A3 is 6 % lower on small cars and 5 % lower on large cars, so depth pretraining does not help small, distant objects.
- **Limitations:** one seed per run; the validation split also selected the checkpoints; continuation is not identical to a fresh 70-epoch schedule (but it was applied to all models alike). A run with a randomly initialised encoder and 3-layer adapters would isolate the value of depth pretraining itself.

Conclusion. A stronger adapter closes about a third of the gap, a randomly initialised decoder changes nothing, and longer training with early stopping shows that the ZipDepth models have converged below YOLO26n. Given the slow and now flat learning progress, the ZipDepth backbone will not reach the original YOLO26n on KITTI under this budget, and it remains 3 to 4 times more expensive.